Unit: 6 Unit 4: Absolute Value
Topic: Absolute Value Function and its Graph

Name: ____________
Date: ____________

Foundation (Jalapeno)

Q1. The graph of $y = |x - 2|$ has a vertex at
(A) (2, 0)
(B) (0, 2)
(C) (2, 2)
(D) (0, 0)
(E) (1, 1)

Q2. The function $y = |x + 3|$ has an axis of symmetry at $x =$
(A) -3
(B) -2
(C) -1
(D) 0
(E) 3

Q3. Which of the following is the graph of $y = 2|x|$?
(A) A wide V-shape
(B) A narrow V-shape
(C) A horizontal line
(D) A vertical line
(E) A parabola

Q4. A spaceship is traveling at a speed of $|v - 200|$ km/h. If the speed is $|v - 200| = 50$ km/h, what is the possible value of v?
(A) 150 km/h
(B) 250 km/h
(C) 100 km/h
(D) 200 km/h
(E) 300 km/h

Q5. The graph of $y = |x| + 2$ is the same as the graph of $y = |x|$ shifted
(A) 2 units up
(B) 2 units down
(C) 2 units left
(D) 2 units right
(E) 1 unit up

Q6. A lab experiment measures the temperature in a container as $|t - 20|$ degrees Celsius. If $|t - 20| = 5$ degrees Celsius, what are the possible values of t?
(A) 15 and 25
(B) 10 and 30
(C) 0 and 40
(D) 5 and 35
(E) 20 and 20

Q7. The function $y = -|x|$ is the reflection of $y = |x|$ across the
(A) x-axis
(B) y-axis
(C) origin
(D) line $y = x$
(E) line $y = -x$

Q8. The planet Mars is traveling at a speed of $|v + 100|$ km/h. If $|v + 100| = 20$ km/h, what are the possible values of v?
(A) -80 and -120
(B) -100 and 0
(C) 0 and 20
(D) 100 and 120
(E) 80 and 100

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 1| + 3$ and identify its vertex, axis of symmetry, and direction of opening. Required calculations and graph should fit in the space provided below.

Q10. The distance of a spaceship from Earth is given by $|d - 500|$ miles. If $|d - 500| = 100$ miles, find the possible values of d and explain the meaning of the absolute value function in this context. Required calculations and explanations should fit in the space provided below.

Chef's Tips

- Tip 1: Encourage students to use graphing calculators or software to visualize the absolute value functions.
- Tip 2: Remind students to consider both positive and negative solutions when working with absolute value equations.
- Tip 3: Emphasize the importance of labeling the vertex, axis of symmetry, and direction of opening when graphing absolute value functions.